

The Methods section

Fish 290

The Methods (or Materials and Methods)

The Methods section is the most straightforward section of the body to write.

The Methods (or Materials and Methods)

Functions of the M&M:

1. Allow replication of the study in the future
 - However, most readers are not interested in replicating your study. What *are* they interested in?
2. Give readers the information they need to contextualize the Results. Each result should have a set of methods attached to it.
3. Allow readers to judge the credibility of the science. Is your approach defensible and appropriate to answer the question?

How to write the Methods

How to order: there is no one-size-fits-all structure of the Methods.

- What does the writer need to know, (and in what order) to be able to understand and judge your results?
- The way you group things will depend on your study
- Should not be strictly chronological: **group like with like**, even if you did things at different times IRL

Still, there are good practices to help readers follow the methods.

- Subheadings are key
- You can even use matching subheadings in the Results section

How to write the Methods

Formatting rules to follow:

1. Use past tense in the Methods
2. Use different font formatting for subheadings
3. Use metric system for all units. Abbreviate units -- 10.5 cm
4. Use military time instead of AM-PM, e.g. 14:30 rather than 2:20 pm
5. When beginning a sentence with a number, always write the number out ("Twenty fish were caught."). In the middle of a sentence, only write out numbers less than ten ("We caught 20 fish.")
6. The level of detail needed depends on your study and the customs of your field. Some details are trivial while others are critical.
7. Spell out abbreviations the first time they are used (except units which are standard)
8. While we're at it, always format all scientific names correctly. Higher taxa are capitalized but not italicized. (Cichlidae). Genera and species are always italicized (*Salmo salar*). After first mention, can abbreviate the genus (*S. salar*).
9. "Fish" = multiple individuals of the same species. "Fishes" = multiple species

Some more hints for writing:

Hint #1: Active vs. Passive voice

Passive:

- Six sites were selected for surveys.
- It was determined that phytoplankton were increasing.

Active:

- We selected six sites for surveys.
- We determined that phytoplankton were increasing.

Describe the difference?

Which is better? There are pros and cons. Choose carefully!

Some more hints for writing:

Hint #1: Active vs. Passive voice

Active:

- Usually shorter and easier to understand- **that's what we want!**
- More engaging: relates the science to the human who did it (you!)
- More specific- we know who did what, the actor is clear
- Can be repetitive

Passive:

- Can be used to draw attention to an object ("Fish were identified to species, but other taxa only to genus").
- Obscures the actor when the actor is irrelevant ("Nonparametric statistics are preferred when data violate assumptions.")
- Sometimes the actor is complex, and sentence is clearer in passive (Compare "Beetles, wasps, and several other insects attacked trees." vs "Trees were attacked by beetles, wasps, and several other insects.")

Some more hints for writing:

Hint #2:

Theatre of mind – painting a picture in somebody else's imagination through words.

Useful when:

- Describing study sites
- Describing experimental design
- Describing an apparatus or complicated experimental tool

Often will accompany a figure

Some more hints for writing:

Hint #2: **Theory of mind**

“Theory of mind” means putting yourself in the reader’s mindset when writing. What would they be thinking? What do they know right now?

Always keep this in mind when writing (any section!). Some things may be obvious to you but not to the reader.

Examples in writing: “It was taken to the lab...” “Temperature affected them...” What is it? Who are them?

Examine Published Methods

- **Look at the de Wit and Johnson 2024 paper on canvas**
- Examine the Materials and Methods
- Note the structure and subheadings
- Note any active or passive voice
- What do you find helpful?
- What would you like to see, but is not there?
- What level of detail is not given, and is that OK?

Assignments for today and for home

- Download the **Methods to Edit 2025** document, and make comments and edits to improve the document
- Attempt to download data for your research project.
If you have issues let me know.
- Start writing your **Introduction**. An outline is a good place to start.
- Read “The Methods Section” on Canvas
- Think about writing your Methods section. You will most likely begin with describing the data you just downloaded, including how it was collected.